Lecture 10

Laplace's equation: separation of variables

©2019–2026 Masayuki Yano. Prepared for ESC384 Partial Differential Equations taught at the University of Toronto.

10.1 Introduction

In this lecture, we seek the solution of Laplace's equation on bounded domains using *separation of variables*. While the technique is fundamentally the same as that considered for the heat equation, we pay particular attention to the extension of the technique to higher spatial dimensions ($n > 1$).

10.2 Laplace's equation on a rectangle

We now consider the solution of Laplace's equation on a rectangle $\Omega \equiv (0, L) \times (0, H)$ using separation of variables:

$$\begin{aligned} -\Delta u &= 0 \quad \text{in } \Omega \equiv (0, L) \times (0, H), \\ u &= g^1 \quad \text{on } \Gamma^1 \equiv \{x = 0\} \times (0, H), \\ u &= g^2 \quad \text{on } \Gamma^2 \equiv \{x = L\} \times (0, H), \\ u &= g^3 \quad \text{on } \Gamma^3 \equiv (0, L) \times \{y = 0\}, \\ u &= g^4 \quad \text{on } \Gamma^4 \equiv (0, L) \times \{y = H\}, \end{aligned} \tag{10.1}$$

where x and y denote the first and second spatial coordinates, respectively, and g^i, $i = 1, \ldots, 4$, are some boundary functions. Note that the boundaries Γ^1, Γ^2, Γ^3, and Γ^4 correspond to the left, right, bottom, and top boundaries, respectively. If the solution is required to be continuous on $\overline{\Omega}$, the boundary functions must agree at the corners; e.g., $g^1(0) = g^3(0)$ at the lower-left corner, with analogous conditions at the other corners.

To solve the boundary value problem (10.1), we again appeal to the principle of superposition. Namely, we decompose the solution u into two parts $u = u^{1,2} + u^{3,4}$, where $u^{1,2}$ is the solution to the boundary value problem with only g^1 and g^2 as the nonhomogeneous boundaries, and $u^{3,4}$ is the

solution to the boundary value problem with only g^3 and g^4 as the nonhomogeneous boundaries. More explicitly, $u^{1,2}$ is the solution to

$$\begin{aligned} -\Delta u^{1,2} &= 0 \quad \text{in } \Omega, \\ u^{1,2} &= g^1 \quad \text{on } \Gamma^1, \\ u^{1,2} &= g^2 \quad \text{on } \Gamma^2, \\ u^{1,2} &= 0 \quad \text{on } \Gamma^3 \cup \Gamma^4; \end{aligned} \tag{10.2}$$

similarly, $u^{3,4}$ is the solution to

$$\begin{aligned} -\Delta u^{3,4} &= 0 \quad \text{in } \Omega, \\ u^{3,4} &= g^3 \quad \text{on } \Gamma^3, \\ u^{3,4} &= g^4 \quad \text{on } \Gamma^4, \\ u^{3,4} &= 0 \quad \text{on } \Gamma^1 \cup \Gamma^2. \end{aligned}$$

We readily see $u = u^{1,2} + u^{3,4}$ by superposition.

We now consider the solution of (10.2) by separation of variables and linear superposition. We recall the two-step procedure:

- *Step 1.* We find a family of separable solutions of the form

  $$u_n(x,y) = X_n(x; C_n, D_n)Y_n(y), \quad n = 1, 2, \ldots,$$

  each of which satisfies the homogeneous boundary conditions on Γ^3 and Γ^4 for any choice of coefficients C_n and D_n, but not (in general) the nonhomogeneous boundary conditions on Γ^1 and Γ^2. (As we have seen for the heat equation, Y_n will be a family of functions in y that satisfy homogeneous boundary conditions at $y = 0$ and $y = H$; hence, we associate the indeterminate coefficients C_n and D_n with functions X_n in x.)

- *Step 2.* We consider a particular choice of coefficients C_n and D_n such that

  $$u^{1,2}(x,y) = \sum_{n=1}^{\infty} X_n(x; C_n, D_n)Y_n(y)$$

  satisfies the nonhomogeneous boundary conditions on Γ^1 and Γ^2. Note that the solution $u^{1,2}$ automatically satisfies the homogeneous boundary conditions on Γ^3 and Γ^4 since each u_n satisfies the homogeneous boundary conditions.

To find a separable solution of the form $u_n(x,y) = X_n(x)Y_n(y)$, we substitute the expression into (10.2). At points where $X_nY_n \neq 0$, we obtain

$$-X_n''Y_n - X_nY_n'' = 0 \quad \Rightarrow \quad \frac{X_n''}{X_n} = -\frac{Y_n''}{Y_n} = \alpha_n$$

for some constant $\alpha_n \in \mathbb{R}$. We have used the fact that X_n''/X_n and Y_n''/Y_n depend on only x and y, respectively, and hence each of them must be a constant.

We now consider the boundary value eigenproblem associated with Y_n; we solve for Y_n (instead of X_n) first because we wish to impose homogeneous boundary conditions at the endpoints $y = 0$ and $y = H$. We first observe that the boundary condition implies that

$$u_n(x, y = 0) = X_n(x)Y_n(0) = 0, \quad \forall x \in (0, L)$$
$$u_n(x, y = H) = X_n(x)Y_n(H) = 0, \quad \forall x \in (0, L).$$

Since we seek a nontrivial solution, we require $X_n \neq 0$ and deduce $Y_n(0) = Y_n(H) = 0$. Our boundary value eigenproblem for Y_n is as follows:

$$-Y_n'' = \alpha_n Y_n \quad \text{in } (0, H),$$
$$Y_n(0) = Y_n(H) = 0.$$

This is a Sturm–Liouville problem. In fact this is a scaled version of the problem considered in Example 4.6, and the eigenpairs are given by

$$Y_n(y) = \sin(\lambda_n y) \quad \text{and} \quad \alpha_n = \lambda_n^2$$

for $\lambda_n = n\pi/H$, $n = 1, 2, \ldots$.

We next consider the ODE associated with X_n:

$$X_n'' = \alpha_n X_n \quad \Rightarrow \quad X_n'' = \lambda_n^2 X_n$$

for $\lambda_n = n\pi/H$. The general solution is given by

$$X_n(x) = C_n \sinh(\lambda_n x) + D_n \cosh(\lambda_n x).$$

We therefore conclude that any solution of the form

$$u^{1,2}(x, y) = \sum_{n=1}^{\infty} (C_n \sinh(\lambda_n x) + D_n \cosh(\lambda_n x)) \sin(\lambda_n y)$$

satisfies Laplace's equation and the associated homogeneous boundary conditions on $y = 0$ and $y = H$.

We now enforce the nonhomogeneous boundary conditions on $x = 0$ and $x = L$. We first enforce $u(x = 0, y) = g^1(y)$ to obtain

$$u^{1,2}(x = 0, y) = \sum_{n=1}^{\infty} D_n \sin(\lambda_n y) = g^1(y)$$

for $\lambda_n = n\pi/H$. We recognize that this is a Fourier sine series and the coefficients are given by

$$D_n = \hat{g}_n^1 \equiv \frac{2}{H} \int_0^H \sin(n\pi y/H) g^1(y)\, dy, \quad n = 1, 2, \ldots. \tag{10.3}$$

We next enforce the boundary condition $u^{1,2}(x = L, y) = g^2(y)$ to obtain

$$u^{1,2}(x = L, y) = \sum_{n=1}^{\infty} (C_n \sinh(\lambda_n L) + \hat{g}_n^1 \cosh(\lambda_n L)) \sin(\lambda_n y) = g^2(y),$$

where we have set $D_n = \hat{g}_n^1$. We note that the "coefficient" $(C_n \sinh(\lambda_n L) + \hat{g}_n^1 \cosh(\lambda_n L))$ must be the n-th Fourier sine coefficient of g^2. Hence

$$C_n \sinh(\lambda_n L) + \hat{g}_n^1 \cosh(\lambda_n L) = \hat{g}_n^2 \equiv \frac{2}{H} \int_0^H \sin(n\pi y/H) g^2(y)\, dy.$$

We rearrange the expression to obtain

$$C_n = \frac{\hat{g}_n^2 - \hat{g}_n^1 \cosh(\lambda_n L)}{\sinh(\lambda_n L)}. \tag{10.4}$$

We therefore conclude that the solution $u^{1,2}$ is given by

$$u^{1,2}(x,y) = \sum_{n=1}^{\infty} \left(\frac{\hat{g}_n^2 - \hat{g}_n^1 \cosh(\lambda_n L)}{\sinh(\lambda_n L)} \sinh(\lambda_n x) + \hat{g}_n^1 \cosh(\lambda_n x) \right) \sin(\lambda_n y).$$

We may group the terms containing $\hat{g}_n^1$ and $\hat{g}_n^2$ to obtain

$$u^{1,2}(x,y) = \sum_{n=1}^{\infty} \left(\hat{g}_n^1 (\cosh(\lambda_n x) - \coth(\lambda_n L) \sinh(\lambda_n x)) + \hat{g}_n^2 \frac{\sinh(\lambda_n x)}{\sinh(\lambda_n L)} \right) \sin(\lambda_n y),$$

which may be further simplified using a hyperbolic identity to yield

$$u^{1,2}(x,y) = \sum_{n=1}^{\infty} \left(\hat{g}_n^1 \frac{\sinh(\lambda_n (L-x))}{\sinh(\lambda_n L)} + \hat{g}_n^2 \frac{\sinh(\lambda_n x)}{\sinh(\lambda_n L)} \right) \sin(\lambda_n y),$$

where $\lambda_n = n\pi/H$.

By the symmetry of the problem, we also find the solution to

$$\begin{aligned} -\Delta u^{3,4} &= 0 \quad \text{in } \Omega, \\ u^{3,4} &= g^3 \quad \text{on } \Gamma^3, \\ u^{3,4} &= g^4 \quad \text{on } \Gamma^4, \\ u^{3,4} &= 0 \quad \text{on } \Gamma^1 \cup \Gamma^2 \end{aligned} \tag{10.5}$$

is

$$u^{3,4}(x,y) = \sum_{n=1}^{\infty} \left(\hat{g}_n^3 \frac{\sinh(\mu_n (H-y))}{\sinh(\mu_n H)} + \hat{g}_n^4 \frac{\sinh(\mu_n y)}{\sinh(\mu_n H)} \right) \sin(\mu_n x),$$

where $\mu_n = n\pi/L$, and $\hat{g}_n^3$ and $\hat{g}_n^4$ are the Fourier sine series coefficients

$$\hat{g}_n^3 \equiv \frac{2}{L} \int_0^L \sin(n\pi x/L) g^3(x)\, dx \quad \text{and} \quad \hat{g}_n^4 \equiv \frac{2}{L} \int_0^L \sin(n\pi x/L) g^4(x)\, dx.$$

It follows that, by linear superposition, the solution to the original problem (10.1) is given by

$$\begin{aligned} u(x,y) &= u^{1,2}(x,y) + u^{3,4}(x,y) \\ &= \sum_{n=1}^{\infty} \left(\hat{g}_n^1 \frac{\sinh(\lambda_n (L-x))}{\sinh(\lambda_n L)} + \hat{g}_n^2 \frac{\sinh(\lambda_n x)}{\sinh(\lambda_n L)} \right) \sin(\lambda_n y) \\ &\quad + \sum_{n=1}^{\infty} \left(\hat{g}_n^3 \frac{\sinh(\mu_n (H-y))}{\sinh(\mu_n H)} + \hat{g}_n^4 \frac{\sinh(\mu_n y)}{\sinh(\mu_n H)} \right) \sin(\mu_n x) \end{aligned}$$

for $\lambda_n = n\pi/H$ and $\mu_n = n\pi/L$.

We make a few observations:

1. The solution is a linear combination of four harmonic functions associated with the four boundary functions $g^1, \ldots, g^4$; i.e., $u = u^1 + u^2 + u^3 + u^4$ where

$$u^1(x,y) = \sum_{n=1}^{\infty} \hat{g}_n^1 \frac{\sinh(\lambda_n(L-x))}{\sinh(\lambda_n L)} \sin(\lambda_n y), \quad u^2(x,y) = \sum_{n=1}^{\infty} \hat{g}_n^2 \frac{\sinh(\lambda_n x)}{\sinh(\lambda_n L)} \sin(\lambda_n y),$$
$$u^3(x,y) = \sum_{n=1}^{\infty} \hat{g}_n^3 \frac{\sinh(\mu_n(H-y))}{\sinh(\mu_n H)} \sin(\mu_n x), \quad u^4(x,y) = \sum_{n=1}^{\infty} \hat{g}_n^4 \frac{\sinh(\mu_n y)}{\sinh(\mu_n H)} \sin(\mu_n x).$$

2. We readily verify that the solution satisfies the boundary conditions:

$$u(x=0,y) = \sum_{n=1}^{\infty} \hat{g}_n^1 \sin(\lambda_n y) = g^1(y), \quad u(x=L,y) = \sum_{n=1}^{\infty} \hat{g}_n^2 \sin(\lambda_n y) = g^2(y),$$
$$u(x,y=0) = \sum_{n=1}^{\infty} \hat{g}_n^3 \sin(\mu_n x) = g^3(x), \quad u(x,y=H) = \sum_{n=1}^{\infty} \hat{g}_n^4 \sin(\mu_n x) = g^4(x).$$

3. Figure 10.1 shows functions $\frac{\sinh(\lambda_n(L-x))}{\sinh(\lambda_n L)}$ and $\frac{\sinh(\lambda_n x)}{\sinh(\lambda_n L)}$ for $H = 1$, which dictate the decay of the n-th mode away from the $x = 0$ and $x = L$ boundaries, respectively. We observe that the functions decay exponentially away from their associated boundaries. For example, away from $x = L$, the ratio $\sinh(\lambda_n x)/\sinh(\lambda_n L)$ behaves approximately as $\exp(-\lambda_n(L-x))$. Moreover, higher modes (i.e., modes associated with larger n) decay more rapidly because $\lambda_n = n\pi/H$.

4. Figure 10.2 shows the solution associated with a nonhomogeneous boundary condition on $x = L$ and with the homogeneous boundary conditions elsewhere: $u_n(x,y) = \hat{g}_n^2 \frac{\sinh(\lambda_n x)}{\sinh(\lambda_n L)} \sin(\lambda_n y)$ for $n = 1$ and 3. We observe that each mode is smooth and decays exponentially away from the boundary.

Example 10.1 (Laplace's equation on a square)**.** We consider Laplace's equation on the unit square $\Omega \equiv (0,1)^2$. The boundary value problem is given by (10.1) for triangular boundary functions

$$g^2(y) = 1 - 2|y - 0.5|, \quad y \in (0,1),$$
$$g^4(x) = 1/2 - |x - 0.5|, \quad x \in (0,1),$$

and $g^1 = g^3 = 0$. To find the solution, we first note that $\lambda_n = \mu_n = n\pi$ since $L = H = 1$. We then find the Fourier sine coefficients of g^2 and g^4:

$$\hat{g}_n^2 = \begin{cases} \frac{8(-1)^{(n-1)/2}}{n^2\pi^2}, & n \text{ odd}, \\ 0, & n \text{ even}, \end{cases}$$
$$\hat{g}_n^4 = \begin{cases} \frac{4(-1)^{(n-1)/2}}{n^2\pi^2}, & n \text{ odd}, \\ 0, & n \text{ even}. \end{cases}$$

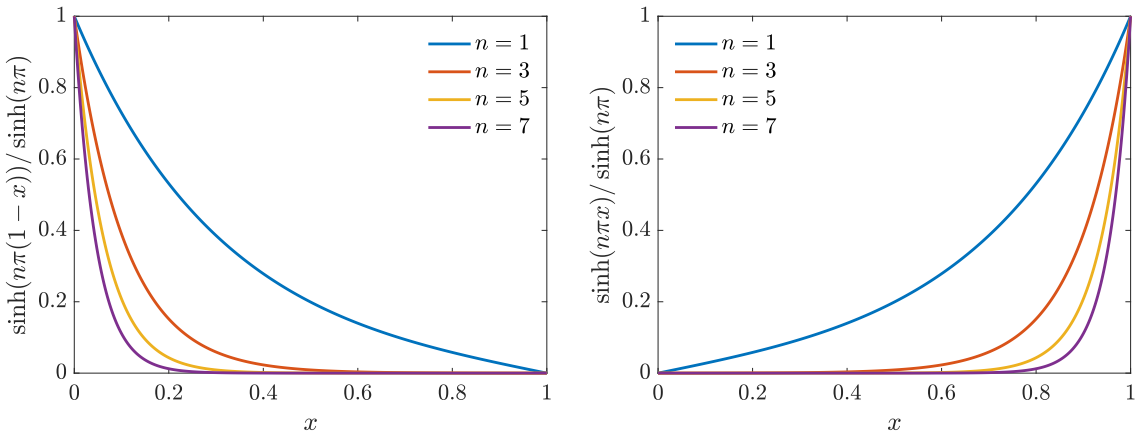
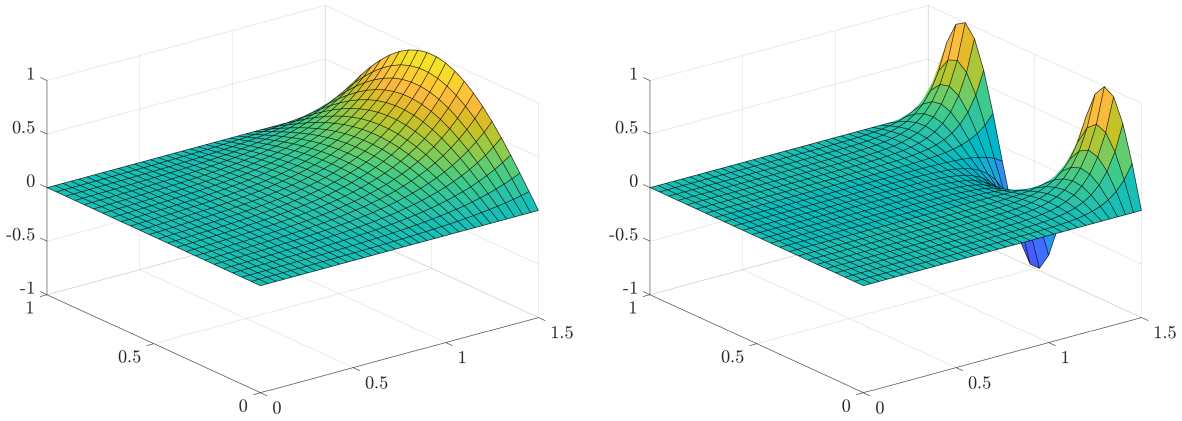


Figure 10.1: Functions $\sinh(n\pi(1-x))/\sinh(n\pi)$ and $\sinh(n\pi x)/\sinh(n\pi)$.



(a) $n = 1$ (b) $n = 3$

Figure 10.2: Solutions to Laplace's equation on $(0, 1.5) \times (0, 1)$ for $g^2(y) = \sin(n\pi y)$ and $g^i = 0$ for $i = 1, 3, 4$.

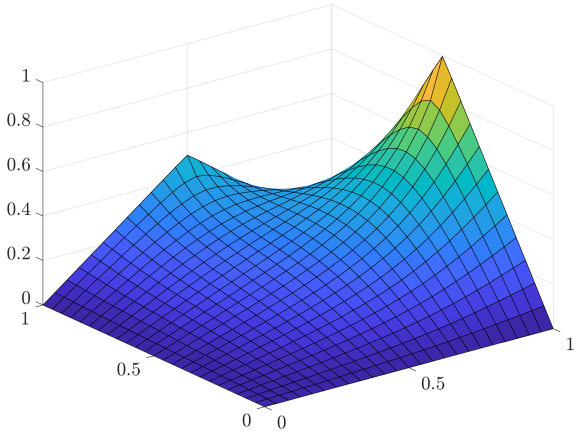


Figure 10.3: Solution to Laplace's equation on a square with triangular boundary functions.

We then substitute the coefficients into our general expression to find

$$u(x,y) = \sum_{n=1,3,5,\ldots}^{\infty} \left(\frac{8(-1)^{(n-1)/2}}{n^2\pi^2} \frac{\sinh(n\pi x)}{\sinh(n\pi)} \sin(n\pi y) + \frac{4(-1)^{(n-1)/2}}{n^2\pi^2} \frac{\sinh(n\pi y)}{\sinh(n\pi)} \sin(n\pi x) \right).$$

Figure 10.3 shows the solution to the problem. We observe that even though the boundary functions are not smooth, the solution is smooth on Ω. In fact it can be shown that harmonic functions are infinitely continuously differentiable in Ω (but not necessarily on $\partial\Omega$). We also observe that the maximum (resp. minimum) principle is clearly satisfied and the maximum (resp. minimum) value is attained on the boundary.

10.3 Laplace's equation on a disk

We now consider the solution of Laplace's equation on a disk $\Omega \equiv \{x \in \mathbb{R}^2 \mid \|x\|_2 < a\}$:

$$\begin{aligned} -\Delta u &= 0 \quad \text{in } \Omega, \\ u &= g \quad \text{on } \partial\Omega. \end{aligned} \tag{10.6}$$

We again use separation of variables but now in polar coordinates. The Laplacian in polar coordinates is

$$\Delta u = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 u}{\partial \theta^2} = \frac{\partial^2 u}{\partial r^2} + \frac{1}{r}\frac{\partial u}{\partial r} + \frac{1}{r^2}\frac{\partial^2 u}{\partial \theta^2} = 0.$$

We substitute the separable form of the solution $u_n(r,\theta) = R_n(r)\Theta_n(\theta)$ and obtain

$$R_n''\Theta_n + \frac{1}{r}R_n'\Theta_n + \frac{1}{r^2}R_n\Theta_n'' = 0.$$

At points where $R_n\Theta_n \neq 0$, we multiply through by $r^2/(R_n\Theta_n)$ to obtain

$$r^2\frac{R_n''}{R_n} + r\frac{R_n'}{R_n} + \frac{\Theta_n''}{\Theta_n} = 0 \quad \Rightarrow \quad r^2\frac{R_n''}{R_n} + r\frac{R_n'}{R_n} = -\frac{\Theta_n''}{\Theta_n} = \alpha_n.$$

Since the left- and right-hand sidess are functions of only r and θ, respectively, we deduce as before that they must be equal to some constant α_n.

We now consider the eigenproblem associated with Θ_n. Because the location of the "cut" that we make in the θ-coordinate is arbitrary, we impose periodic boundary conditions. In addition, we solve for Θ_n (instead of R_n) first because the periodic boundary condition can be readily incorporated into the separable form of the solution, just like the homogeneous boundary conditions in the rectangular case. The resulting eigenproblem is given by

$$\begin{aligned} -\Theta_n'' &= \alpha_n \Theta_n \quad \text{in } (0, 2\pi), \\ \Theta_n(0) &= \Theta_n(2\pi), \\ \Theta_n'(0) &= \Theta_n'(2\pi), \end{aligned}$$

This is a periodic Sturm–Liouville problem. The general solution to the ODE is given by

$$\Theta_n(\theta) = A_n \cos(\lambda_n \theta) + B_n \sin(\lambda_n \theta)$$

for $\lambda_n = \sqrt{\alpha_n}$. We then impose the periodic boundary condition to find that $\lambda_n = n$, $n = 0, 1, 2, \ldots$, and hence

$$\Theta_n(\theta) = A_n \cos(n\theta) + B_n \sin(n\theta),$$

and $\alpha_n = \lambda_n^2 = n^2$. (Note that $\Theta_{n=0} = A_n$ is a constant function, which of course is periodic.)

Having found $\alpha_n = n^2$, we next observe that the ODE for R_n must satisfy

$$r^2 R_n'' + r R_n' - n^2 R_n = 0.$$

This is a Cauchy–Euler equation of order two, whose solution is of the form $R_n(r) = r^m$ for some m. We substitute the form of the solution to obtain

$$m(m-1)r^m + mr^m - n^2 r^m = 0 \quad \Rightarrow \quad m^2 - n^2 = 0. \quad \Rightarrow \quad m = \pm n.$$

It follows that $R_n(r) = C_n r^n + D_n r^{-n}$ if $n \neq 0$. For $n = 0$, the solution is given by $R_{n=0}(r) = C_{n=0} + D_{n=0} \log(r)$. We now impose the condition that R_n should be bounded for $0 \leq r < a$ and in particular at the origin; we hence reject the solution r^{-n} and $\log(r)$ which are unbounded at the origin. It follows that our solution is

$$R_n(r) = C_n r^n, \quad n = 0, 1, 2, \ldots.$$

Hence our family of separable solutions is

$$u_n(r, \theta) = R_n(r)\Theta_n(\theta) = r^n (A_n \cos(n\theta) + B_n \sin(n\theta)), \quad n = 0, 1, 2, \ldots.$$

We appeal to the principle of superposition and rescale the constant term by $1/2$ and all other terms by $1/a^n$ for convenience to obtain

$$u(r, \theta) = \frac{1}{2} A_0 + \sum_{n=1}^{\infty} \left(\frac{r}{a}\right)^n (A_n \cos(n\theta) + B_n \sin(n\theta)). \tag{10.7}$$

To find the coefficients $(A_n)_{n=0}^{\infty}$ and $(B_n)_{n=1}^{\infty}$, we apply the boundary condition at $r = a$:

$$u(r = a, \theta) = \frac{1}{2} A_0 + \sum_{n=1}^{\infty} (A_n \cos(n\theta) + B_n \sin(n\theta)) = g(\theta);$$

this is the Fourier series for g with period 2π and hence the coefficients are given by

$$A_n = \frac{1}{\pi}\int_{-\pi}^{\pi} g(\theta)\cos(n\theta)\,d\theta, \quad n = 0, 1, \ldots, \tag{10.8}$$
$$B_n = \frac{1}{\pi}\int_{-\pi}^{\pi} g(\theta)\sin(n\theta)\,d\theta, \quad n = 1, 2, \ldots.$$

Hence the Fourier series representation of the solution to (10.6) is given by (10.7) with coefficients (10.8). We make a few observations:

1. For $n \geq 1$, the amplitude of each mode depends on $(r/a)^n$ with $r/a \leq 1$, and high modes—modes with a large n that exhibit rapid oscillation on the boundary—decay rapidly away from the boundary. The $n = 0$ mode is constant and does not decay. This behavior is shown in Figure 10.4 and is similar to the decay away from the boundary observed for rectangular domains.

2. The solution (as expected) satisfies the mean-value formulas over Ω and $\partial\Omega$. The value at the center of the disk is
$$u(r = 0, \theta) = \frac{1}{2}A_0.$$
The average of the solution over the disk is
$$\begin{aligned}\frac{1}{\pi a^2}\int_\Omega u dx &= \frac{1}{\pi a^2}\int_{\theta=-\pi}^{\pi}\int_{r=0}^{a}\left(\frac{1}{2}A_0 + \sum_{n=1}^{\infty}\left(\frac{r}{a}\right)^n (A_n\cos(n\theta) + B_n\sin(n\theta))\right) r dr d\theta \\ &= \frac{1}{\pi a^2}\int_{\theta=-\pi}^{\pi}\int_{r=0}^{a}\frac{1}{2}A_0 r dr d\theta = \frac{1}{2}A_0,\end{aligned}$$
where the second equality follows because all of the sine and cosine integrals vanish when integrated over $(-\pi, \pi)$. The average over the circumference is
$$\begin{aligned}\frac{1}{2\pi a}\int_{\partial\Omega} u ds &= \frac{1}{2\pi a}\int_{\theta=-\pi}^{\pi} u(r = a, \theta) a d\theta \\ &= \frac{1}{2\pi a}\int_{\theta=-\pi}^{\pi}\left(\frac{1}{2}A_0 + \sum_{n=1}^{\infty}(A_n\cos(n\theta) + B_n\sin(n\theta))\right) a d\theta \\ &= \frac{1}{2\pi a}\int_{\theta=-\pi}^{\pi}\frac{1}{2}A_0 a d\theta = \frac{1}{2}A_0,\end{aligned}$$
where the third equality again follows because all of the sine and cosine integrals vanish. Hence, we observe that, for any boundary condition g, the value of the solution at the center of the disk is equal to (i) its average over the disk and (ii) its average over the circumference, as expected from the mean-value formulas.

Example 10.2 (Laplace's equation on a disk)**.** Let $\Omega \equiv \{x \in \mathbb{R}^2 \mid \|x\|_2 < 2\}$ and consider
$$\begin{aligned}-\Delta u &= 0 \quad \text{in } \Omega, \\ u(r = 2, \theta) &= 1 + \cos(3\theta), \quad \theta \in [-\pi, \pi).\end{aligned}$$

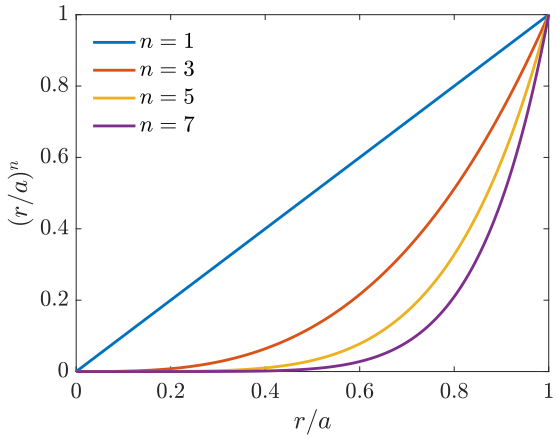


Figure 10.4: Function $(r/a)^n$.

We wish to find the solution to the boundary value problem. We could find the Fourier coefficients using the formula (10.8), but for this simple boundary condition we can find the coefficients by inspection. Namely, we wish to find $(A_n)_{n=0}^{\infty}$ and $(B_n)_{n=1}^{\infty}$ such that

$$u(r = 2, \theta) = \frac{1}{2}A_0 + \sum_{n=1}^{\infty}(A_n \cos(n\theta) + B_n \sin(n\theta)) = 1 + \cos(3\theta).$$

We match the coefficients on the left- and right-hand sides to observe that $A_0/2 = 1$ (i.e., $A_0 = 2$), $A_3 = 1$, and all other coefficients should be zero. Hence the solution is

$$u(r, \theta) = 1 + \left(\frac{r}{2}\right)^3 \cos(3\theta).$$

Figure 10.5 shows the solution. We observe, as expected, the solution is smooth. In addition, we readily verify the (weak) maximum and minimum principles: the maximum and minimum of 2 and 0, respectively, occur on the boundary. The strong maximum and minimum principles are also satisfied: the solution strictly inside the domain Ω (i.e., not on the boundary) are all strictly greater than 0 and strictly less than 2, since $(r/2)^3 < 1$ for all $r < 2$. Finally, we observe that the mean-value formulas are satisfied; the value at the center, the average over the disk, and the average over the circumference are all equal to 1.

10.4 Summary

We summarize key points of this lecture:

1. Separation of variables can be applied to solve Laplace's equation in two (or higher) dimensions.

2. The solution to Laplace's equation on a rectangle or disk can be expressed in terms of the Fourier coefficients of the boundary data.

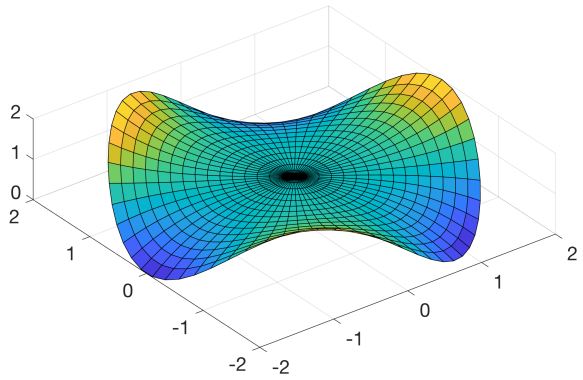


Figure 10.5: Solution to Laplace's equation on a disk.

3. Nonconstant boundary modes decay away from the boundary on a rectangle or disk. Higher-frequency modes decay more rapidly; i.e., the effects of highly oscillatory boundary modes are confined to a thin region near the boundary.